Roman Garipov

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Research Interests

LLM Reasoning, Reinforcement Learning, Looped Models, Latent Reasoning, AI Safety

Education

New York University (NYU), New York, NY, USA *Sep 2026 – Present*
PhD Student in Computer Science
Advisors: Prof. Pavel Izmailov and Prof. Pratyusha Sharma

ITMO University, Saint Petersburg, Russia *Sep 2023 – Jun 2025*
Master of Science in Programming and Artificial Intelligence *GPA: 4.94 /5.00*
Thesis: *Accelerating Large Language Model Inference Using Speculative Decoding*
Relevant Courses: Natural Language Processing, Reinforcement Learning, Generative Artificial Intelligence, Advanced Machine Learning, Deep Learning

ITMO University, Saint Petersburg, Russia *Sep 2019 – Jun 2023*
Bachelor of Science in Applied Mathematics and Programming
Thesis: *Designing a Deep Learning Architecture to Process Tabular Data for Meta-Learning*
Relevant Courses: Calculus, Linear Algebra, Discrete Mathematics, Probability Theory, Statistics, Data Analysis, Machine Learning, Algorithms and Data Structures

Publications

Reasoning Shift: How Context Silently Shortens LLM Reasoning **COLM 2026, ER Workshop Spotlight**
Gleb Rodionov, **Roman Garipov**, George Yakushev

Description: Studied how reasoning LLMs silently shorten reasoning traces and reduce self-verification when solving the same problem under distracting or long-context conditions. Demonstrated significant reductions in reasoning length (up to 65%) and showed that targeted supervised fine-tuning can improve robustness to irrelevant context.

arxiv.org/abs/2604.01161

AutoJudge: Task-Specific Lossy Speculative Decoding for Faster LLM Inference **NeurIPS 2025, Poster**
Roman Garipov, Fedor Velikonivtsev, Ivan Ermakov, Ruslan Svirschevski, Vage Egiazarian, Max Ryabinin

Description: Proposed a task-specific lossy speculative decoding framework for accelerating LLM inference by selectively skipping unimportant token verifications. Achieved up to 2× speedup with minimal accuracy loss on reasoning (GSM8K) and code generation (LiveCodeBench) tasks, demonstrating strong generalization and efficiency gains. The method shows similar performance gains when implemented in the vLLM inference framework.

arxiv.org/abs/2504.20039 garipovroma/autojudge

Hogwild! Inference: Collaborative LLM Reasoning via Shared Attention Cache **NeurIPS 2025, Spotlight**
Gleb Rodionov, **Roman Garipov**, Alina Shutova, George Yakushev, Erik Schultheis, Vage Egiazarian, Anton Sinitsin, Denis Kuznedelev, Dan Alistarh

Description: Proposed a parallel LLM inference framework that expands batch size to simulate multiple collaborating workers sharing a concurrently updated attention cache. Demonstrated that a single modern LLM can effectively emulate multi-agent reasoning without fine-tuning, surpassing self-consistency and other explicit reasoning techniques. The approach improves hardware utilization and enables efficient, task-adaptive cooperation strategies using RoPE-based position handling.

arxiv.org/abs/2504.06261 eqimp/hogwild_llm

Research Experience

Yandex Research x HSE Laboratory Moscow, Russia
ML Research Resident, LLM research group *May 2024 – August 2026*

- Conducted full-time research on LLM efficiency, reasoning, and reinforcement learning, resulting in three papers (see Publications).
- Proposed and implemented a solution for one-click on-demand Jupyter Lab sessions with configurable resources within the internal infrastructure.
- Developed a lightweight framework for parallel GPU computations to accelerate large-scale LLM inference across multiple GPUs: `yandexdataschool/gpu_parallel`, now actively used by the entire research group.
- Delivered paper reviews presentations at internal LLM reading groups and Yandex seminars.
- Supervised two undergraduate theses on reasoning and speculative decoding.

Work Experience

OneMarketData Yerevan, Armenia
Data Scientist *Jun 2023 – May 2024*

- Applied classical ML algorithms using scikit-learn and CatBoost for market impact prediction.
- Contributed to an automatic market surveillance assistant based on statistical methods, using LLMs via LangChain to reduce manual effort in filtering alerts.

VK (VKontakte) Saint Petersburg, Russia
Junior Machine Learning Engineer, Recommender Systems Team *Jul 2022 – May 2023*

- Designed and integrated large-scale data pipelines (100K+ users) using Apache Spark to collect user interaction data and prepare training datasets.
- Implemented a pipeline for automatic model retraining as new data arrives, ensuring recommendations stay up-to-date; neural models were implemented with PyTorch.
- Adapted research-driven techniques to reduce popularity bias, improving both diversity and user engagement.

ITMO University ML Lab Saint Petersburg, Russia
Deep Learning Intern *Mar 2022 – May 2022*

- Collected and labeled a custom dataset (500+ images) for semantic line segmentation.
- Implemented baseline methods using OpenCV and a ResNet-based deep learning model in PyTorch.
- Applied the Deep Hough Transform technique, significantly improving baseline performance to a production-viable level.
- `garipovroma/conveyor-detection`

Yandex.Market Saint Petersburg, Russia
Software Engineer Intern *Jul 2021 – Oct 2021*

- Built and optimized backend features in Java (Spring), integrated internal APIs, and improved database performance for key user-facing services.

Teaching Experience

HSE University Moscow, Russia
Instructor, Bachelor Deep Learning Research Seminar *Academic Year 2025–2026*

- Reviewed and provided feedback on student presentations of notable deep learning papers, led paper discussions, and prepared test questions

Yandex School of Data Analysis Online
Teaching Assistant, NLP Course — For You (GitHub) *Fall Semester 2025*

- Reviewed and graded students' NLP assignments; provided feedback and guidance on both implementation and theoretical aspects

ITMO University Online
Lecturer, Master's Course on Modern Generative Modeling (organized by Arip Asadulaev) *Fall Semester 2025*

- Prepared and delivered three lectures on language modeling for master's students, covering language modeling basics, autoregressive modeling, masked diffusion methods, and the current state of LLMs (training stages, inference, reasoning)

Innopolis University Summer School Innopolis, Russia
Instructor, Algorithms and Data Structures (Competitive Programming) *July 2020*

- Taught algorithms and data structures to high school students preparing for Olympiad-level programming competitions
- Designed lectures and problem sets, reviewed student code, and provided mentorship on debugging and optimization techniques

Awards

Master's Portfolio Contest (ITMO University) – *Winner*, admitted without entrance exams *July 2023, Saint Petersburg, Russia*
Evraz AI Data Science Hackathon – *Awardee* *Nov 2021, Online, Russia*
Serokell.io Scholarship – *Recipient*, awarded to outstanding ITMO students *Jan 2021, Saint Petersburg*
SPbSU Olympiad in Competitive Programming – *Awardee* *Feb 2019, Saint Petersburg, Russia*
ITMO University Olympiad in Informatics – *Awardee* (2018, 2019) *Feb 2018/2019, Saint Petersburg, Russia*
Regional Olympiad in Competitive Programming – *Awardee* *Jan 2019, Innopolis, Russia*

Skills

- **Programming Languages:** Python, C++, Java
- **DL and LLM:** PyTorch, transformers, accelerate, trl, verl, vLLM
- **Data Processing and Visualization:** NumPy, Pandas, Spark, Hadoop, Matplotlib
- **Other:** LaTeX, Git, Linux